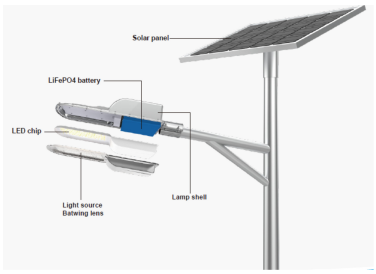
SERIES TYPE STREETLIGHTS

SP / SY / SH SERIES DESIGN TY





SP / SY / SH Solar Streetlight System

Product List

Series	Models
SY Series	SY-80, SY-100
SH Series	SH-100-M, SH-120-M
SP Series	SP-150, SP-200-A, SP-200

Key Features

- Separate solar panel and lamp assembly.
- Monocrystalline silicon solar cells.
- Long-life $LiFePO_4$ battery.
- Patented bat-wing lens.
- IPC 5.0 intelligent power control.
- Weather-responsive discharge management.
- Battery, charging and load indicators.
- Software and hardware battery protection.
- Remote-control lighting modes.

Installation

Mounting Height	5–10 m, depending on model
Pole-Top Diameter	60 mm
Installation Spacing	20–40 m

Lighting Performance

LED Power Range	20–70 W
Luminous Flux	3,400–12,000 lm
System Efficacy	Up to 160 lm/W
Colour Temperature	6000–7500 K
Colour Rendering	Ra ≥ 70
Beam Options	$150° \times 70°$ $155° \times 55°$ $160° \times 45°$
LED Lifespan	50,000 hours
Protection Rating	IP65
Operating Temperature	$-20\,°C$ to $+60\,°C$

Solar Panel & Battery

Panel Type	Monocrystalline silicon
Panel Power Range	60–200 W
Panel Voltage	6 V / 18 V
Solar Cell Efficiency	17–20%
Battery Type	$LiFePO_4$
Battery Energy	192–640 Wh
Battery Cycle Life	More than 2,000 cycles
Rainy-Day Operation	5–7 days, brochure claim

System Design

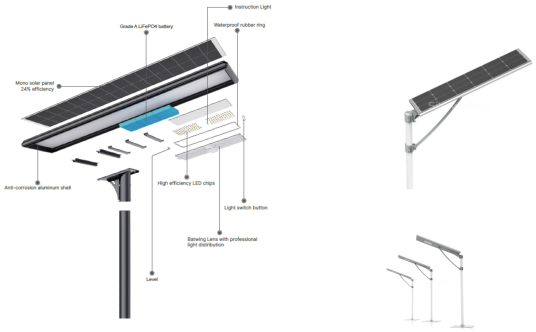
The all-in-two configuration separates the solar panel from the lighting assembly, simplifying transport, installation and maintenance.

Technical note: Values are series-wide ranges. The extracted brochure table does not clearly preserve every model-to-specification relationship. Confirm individual model configurations before ordering. Model numbers are not LED wattage ratings. Rainy-day operation depends on lighting mode, battery state and solar conditions.

AIO TYPE STREETLIGHTS

INTEGRATED SOLAR LIGHTING

GLOBAL SUNTECH
ITS and Power Systems



All-in-one type

Key Features

- Integrated solar lighting assembly with a high-efficiency panel.
- Grade A $LiFePO_4$ battery.
- Multi-peak MPPT charging technology.
- IPC 5.0 intelligent power management.
- Modular battery and LED components.
- Anti-corrosion aluminum housing.
- Battery capacity indicators.
- Remote-selectable lighting modes.

Lighting & Protection

System Efficacy	Greater than 160 lm/W
Colour Temperature	3000–7500 K
Colour Rendering	Ra ≥ 70
LED Lifespan	50,000 hours
Protection Rating	IP65
Operating Temperature	$-20\,^{\circ}\mathrm{C}$ to $+60\,^{\circ}\mathrm{C}$
Charging Temperature	Above $0\,^{\circ}\mathrm{C}$

Solar & Battery System

Solar Panel Type	Monocrystalline silicon
Solar Cell Efficiency	22–24%
Panel Power Range	42–150 W
Battery Chemistry	Grade A $LiFePO_4$
Battery Energy Range	144–960 Wh
Battery Capacity Range	45–300 Ah
Installation Spacing	20–40 m

Operating Modes

DEMO	Light on for one minute for testing
OFF	Light off, day or night
L Mode	100% for 2 h; 70% for 2 h; 30% until dawn
T Mode	100% for 3 h; 50% for 2 h; 20% for 6 h; 50% for 1 h before dawn
M Mode	Whole-night operation at reduced output power
U Mode	Highlight mode; timed output boost
Default Mode	T mode

Battery Management

The brochure describes software and hardware protection, current equalisation, and intelligent discharge planning. Battery capacity retention is stated as 90% after 2,000 cycles under the manufacturer's test conditions.

Series Type Product List

Model	Solar Panel	Luminous Flux	Battery	Mounting Height
AIO-40	42 W	4,000 lm	144 Wh / 45 Ah	4–6 m
AIO-60	60 W	6,000 lm	240 Wh / 75 Ah	6–7 m
AIO-80	90 W	10,000 lm	384 Wh / 120 Ah	6–8 m
AIO-120	120 W	15,000 lm	640 Wh / 200 Ah	8–10 m
AIO-150	150 W	18,000 lm	960 Wh / 300 Ah	10–12 m

F3 LED FLOODLIGHTS

AC-POWERED AREA LIGHTING — MODEL ST-F3-400W



F3 High-Output LED Floodlight

Key Features

- Universal 100–277 V AC input.
- Die-cast ADC12 aluminium housing.
- IP66 dust and water protection.
- Multiple narrow and wide beam options.
- High-power-factor LED driver.
- 20 kV surge protection stated in key features.
- Optional dimming and sensor controls.
- Suitable for outdoor wet locations.

Applications

- Roads and large outdoor areas.
- Billboards and building illumination.
- Tunnels, bridges and culverts.
- Public squares and general area lighting.

Optional Controls

Dimming	0–10 V / PWM / DALI
Sensors	Photocell / microwave motion sensor
NEMA Receptacle	Optional 3-pin, 5-pin or 7-pin
LED Brands	San'an standard; Osram, Philips and others optional

Electrical & Construction

Model	ST-F3-480W
Input Voltage	100–277 V AC
Frequency	50 / 60 Hz
Power Supply Efficiency	Greater than 92%
Power Factor	≥ 0.95
THD	$\leq 10\%$ at 220–240 V AC
Protection Rating	IP66
Housing Material	Die-cast ADC12 aluminium
Net Weight	15 kg; 17 kg with hood cover
Operating Temperature	$-40\,^{\circ}\mathrm{C}$ to $+60\,^{\circ}\mathrm{C}$
Humidity Range	10–90% RH

Lighting Performance

Efficacy Options	110–160 lm/W
Listed Luminous Flux	55,000–80,000 lm, $\pm 5\%$
Colour Temperature	2700–5700 K, adjustable as listed
Colour Rendering	Ra > 70
LED Package Options	3535 / 3030 / 5050
Beam Options	$7°$, $15°$, $30°$, $60°$, $90°$, $120°$
Road Optics	T2M / T3M / T4M, configuration-dependent
Listed Lifetime	More than 50,000 hours

Optical Configurations

Efficacy	Luminous Flux	LED Package	Lens Type
110 lm/W	55,000 lm $\pm 5\%$	3535	4-in-one
120 lm/W	60,000 lm $\pm 5\%$	3030	12-in-one
140 lm/W	70,000 lm $\pm 5\%$	3030	16-in-one
150 lm/W	75,000 lm $\pm 5\%$	5050	4-in-one
160 lm/W	80,000 lm $\pm 5\%$	3030	24-in-one

Technical note: This is an AC-powered floodlight, not a solar streetlight. The source identifies the model as ST-F3-480W, but its listed flux and efficacy combinations imply approximately 500 W; confirm rated input power. The brochure also lists both 3-year and 5-year warranties. Confirm the applicable warranty, optical configuration and electrical ratings before ordering.